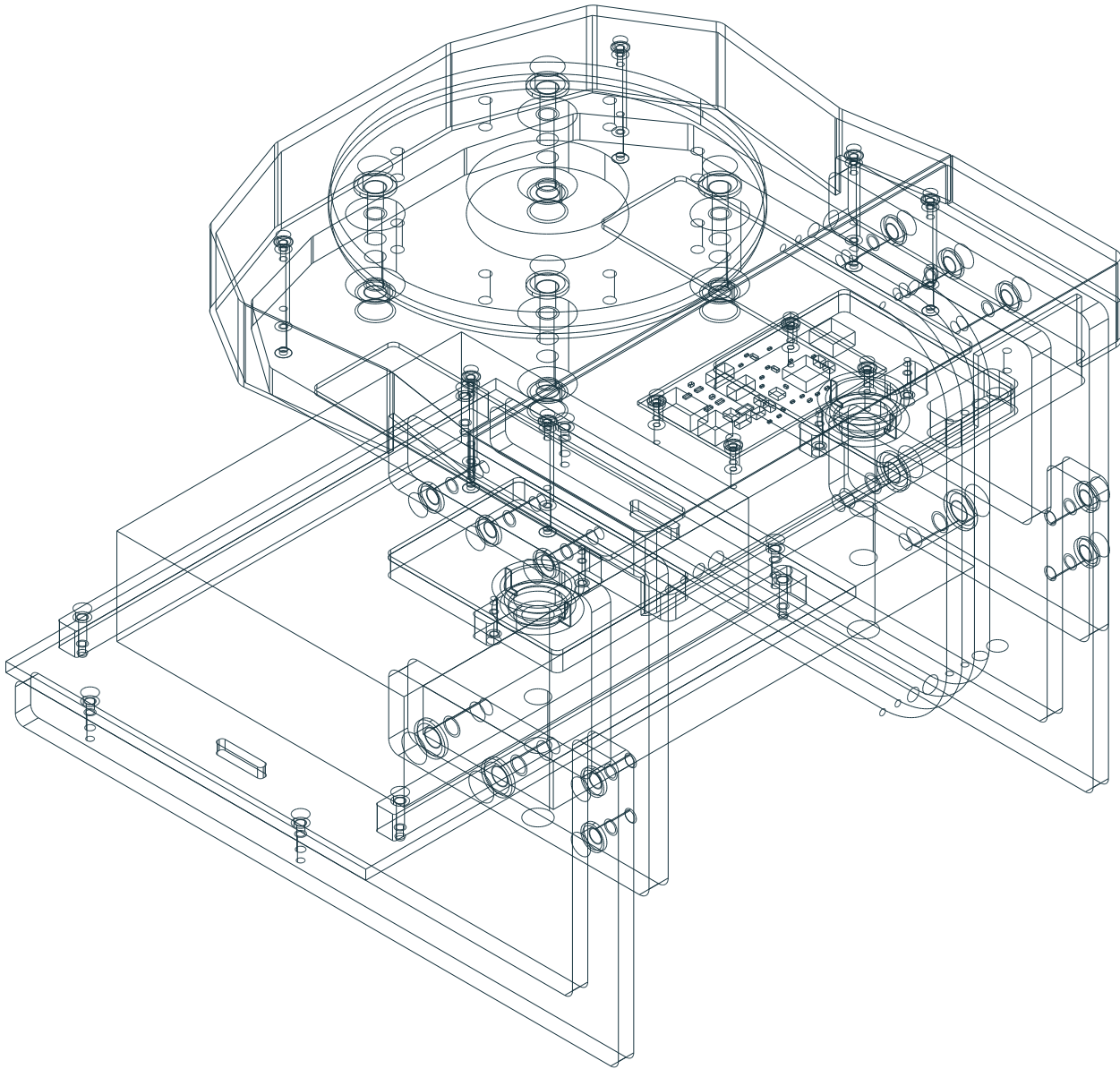


真实 STEP 重读 / 全局坐标 X 左右、+Y 向桌内、+Z 向上 / 原创零件名义几何

名义装配 / 同一全局坐标 比例 0.4:1



原点与接口须在加工前共同确认。紧固扭矩、公差等级、材料状态、线束弯曲半径和压接规范若未在原始清单中给出，均保持待确认。

manifest SHA256 ad065c979d159c1636a3b4f45636eb2d2e782b9553024c0c5ffa4498b730d9ef

本图册的用途

以实体投影和特征表支持零件报价、工艺及装配审查。尺寸由 STEP 或 manifest 读取，不从概念图估算。

读图约定

曲边按 0.01 mm
弦偏差采样；边线不消隐。各正交视图注明坐标平面与比例，不按三角法排图。

基准与制造状态

采用同一装配全局原点。各零件给出全局极值；manifest 中 local_bbox 按原字段保留。未定义的公差、表面处理、有效牙深及供应商接口不得由本图补定。

实体不等于总装通过

PCB、插头、线缆和采购件是否进入实体须看 manifest。简化包络不作为真实供应商模型。尺寸表不提供承载或电气验收结论。

仅列 manifest 中带加工 STEP 的零件；采购件、紧固件和 PCB 另见对应清单

特征 / 属性路径	数值、方向和限定说明（长度默认 mm；其他单位以字段名为准）
B04-101-DECK	数量 1 材料 6061-T651 工艺 CNC profile + mill + drill/tap B04-101-DECK.step
B04-102-CPLATE-L	数量 1 材料 S355 工艺 Profile cut oversize; finish mill + drill/tap B04-102-CPLATE-L.step
B04-102-CPLATE-R	数量 1 材料 S355 工艺 Profile cut oversize; finish mill + drill/tap B04-102-CPLATE-R.step
B04-103-BRIDGE	数量 1 材料 6061-T651 工艺 CNC mill + drill/tap B04-103-BRIDGE.step
B04-104-FLANGE	数量 1 材料 6061-T651 工艺 CNC turn/mill + drill/tap B04-104-FLANGE.step
B04-105-POST-1	数量 1 材料 6061-T651 工艺 Turn + drill/tap B04-105-POST-1.step
B04-105-POST-2	数量 1 材料 6061-T651 工艺 Turn + drill/tap B04-105-POST-2.step
B04-105-POST-3	数量 1 材料 6061-T651 工艺 Turn + drill/tap B04-105-POST-3.step
B04-105-POST-4	数量 1 材料 6061-T651 工艺 Turn + drill/tap B04-105-POST-4.step
B04-106-TOPPAD--1	数量 1 材料 silicone 工艺 Knife/waterjet cut B04-106-TOPPAD--1.step
B04-107-PRESSPAD--1	数量 1 材料 6061-T651 工艺 CNC mill/drill/tap B04-107-PRESSPAD--1.step
B04-108-LOWPAD--1	数量 1 材料 silicone 工艺 Knife/waterjet cut B04-108-LOWPAD--1.step
B04-109-FOOT-CAGE--1--1	数量 1 材料 PETG 工艺 FDM; 0.2 layer, 4 walls, solid around screws B04-109-FOOT-CAGE--1--1.step
B04-109-FOOT-CAGE--1-1	数量 1 材料 PETG 工艺 FDM; 0.2 layer, 4 walls, solid around screws B04-109-FOOT-CAGE--1-1.step
B04-106-TOPPAD-1	数量 1 材料 silicone 工艺 Knife/waterjet cut B04-106-TOPPAD-1.step
B04-107-PRESSPAD-1	数量 1 材料 6061-T651 工艺 CNC mill/drill/tap B04-107-PRESSPAD-1.step
B04-108-LOWPAD-1	数量 1 材料 silicone 工艺 Knife/waterjet cut B04-108-LOWPAD-1.step
B04-109-FOOT-CAGE-1--1	数量 1 材料 PETG 工艺 FDM; 0.2 layer, 4 walls, solid around screws B04-109-FOOT-CAGE-1--1.step
B04-109-FOOT-CAGE-1-1	数量 1 材料 PETG 工艺 FDM; 0.2 layer, 4 walls, solid around screws B04-109-FOOT-CAGE-1-1.step
B04-201-CRADLE-L	数量 1 材料 6061-T651 工艺 CNC profile + drill/tap B04-201-CRADLE-L.step
B04-201-CRADLE-R	数量 1 材料 6061-T651 工艺 CNC profile + drill/tap B04-201-CRADLE-R.step
B04-202-TRAY	数量 1 材料 6061-T651 工艺 CNC profile + drill/tap B04-202-TRAY.step
B04-203-STOP-REAR	数量 1 材料 6061-T651 工艺 Mill + drill B04-203-STOP-REAR.step
B04-203-STOP-FRONT	数量 1 材料 6061-T651 工艺 Mill + drill B04-203-STOP-FRONT.step
B04-301-COVER-REAR	数量 1 材料 PETG 工艺 FDM; 0.2mm layer; 4 walls; supports under roof B04-301-COVER-REAR.step
B04-301-COVER-FRONT	数量 1 材料 PETG 工艺 FDM; 0.2mm layer; 4 walls; supports under roof B04-301-COVER-FRONT.step

manifest SHA256 ad065c979d159c1636a3b4f45636eb2d2e782b9553024c0c5ffa4498b730d9ef

逐字沿用参数声明；孔位置公差的具体基准与检验方法仍需制造评审

特征 / 属性路径	数值、方向和限定说明（ 长度默认 mm；其他单位以字段名为准 ）
coordinates.x	right
coordinates.y	inward from desk rear edge
coordinates.z	up from desk top
tolerances.machined_unless_noted_mm	0.2
tolerances.mounting_hole_position_mm	0.1
tolerances.printed_unless_noted_mm	0.5
tolerances.mating_face_flatness_mm	0.1
notes	[All tapped holes use pilot cylinders in STEP, with complete thread callouts in feature tables., Base-only interface is new; arm motor adapter is a later module., 48 mm rear design clearance is a prototype allocation, not a measured installation minimum., No load-bearing printed components; never test the full arm on an unqualified desktop.]
B04-101-DECK	Profile R4; metal top z14, desk pad plane z0. Side tapped holes drill 17 deep, full thread >=14. Bottom flange-post counterbores seat at z10.5; screw heads remain above z2.5.
B04-102-CPLATE-L, B04-102-CPLATE-R	All profile corners R5. 10 mm certified steel plate; coat after machining; mask mating faces. Left/right differ by installation only; hole axes shown from outer face.
B04-103-BRIDGE	M12 thrust screws: certified DIN6332-5.8, AF6 drive; no ordinary cup-point grub screws.
B04-104-FLANGE	Base test interface only. OD160 / ID60, PCD120: 8 x M6 phase22.5deg. Face z58 flatness0.1; no commercial motor compatibility claimed.
B04-105-POST-1, B04-105-POST-2, B04-105-POST-3, B04-105-POST-4	OD22 x32; M8x1.25-6H through; end faces parallel0.05. Four identical posts.
B04-106-TOPPAD--1, B04-106-TOPPAD-1	2 mm solid silicone sheet 60 Shore A nominal; adhesive to metal only; friction not qualified.
B04-107-PRESSPAD--1, B04-107-PRESSPAD-1	Steel thrust foot directly bears on flat underside; polymer retaining clips carry no clamp load.
B04-108-LOWPAD--1, B04-108-LOWPAD-1	2 mm solid silicone sheet; bond to spreader upper face.
B04-109-FOOT-CAGE--1--1, B04-109-FOOT-CAGE--1-1, B04-109-FOOT-CAGE-1--1, B04-109-FOOT-CAGE-1-1	Split retainer, D28 x8 chamber / D22 throat; 0.4mm split gap. Non-load-bearing. Post-drill screw holes D3.4 +0.2/0. Steel foot D25 flange x6, D18 neck x7 allocation. Verify received part articulation before use.
B04-201-CRADLE-L, B04-201-CRADLE-R	Profile corners R4. Withdraw controller toward +Y; cradle removed without loosening main desk clamp.
B04-202-TRAY	Controller mechanical gauge 180x150x50, max3kg allocation; controller ventilation to be checked separately. Do not install screws within the central controller footprint.
B04-203-STOP-REAR, B04-203-STOP-FRONT	Front stop removable using 2 M4x16; controller gauge slides toward +Y. Use separate adjustable strap through tray side slots to retain vertically.
B04-301-COVER-REAR, B04-301-COVER-FRONT	Nominal wall2.5, roof3.0; clearances checked on nominal geometry. Print as separate front/rear pieces; rear lifts vertically for PCB access; seam0.8mm. Cosmetic shroud carries no arm or clamp load. Post-drill screw holes D3.4 +0.2/0 at assembly jig.

manifest SHA256 ad065c979d159c1636a3b4f45636eb2d2e782b9553024c0c5ffa4498b730d9ef

相同标注已成组；全部入口坐标保留 / 本表不含采购包络的伪加工尺寸

特征 / 属性路径	数值、方向和限定说明 (长度默认 mm ; 其他单位以字段名为准)
B04-101-DECK D8.5 THRU	axis=[0, 0, 1], model_depth_mm=12, model_diameter_mm=8.5 全局入口坐标 XYZ: (-45, 90, 2); (45, 90, 2); (-45, 180, 2); (45, 180, 2)
B04-101-DECK CBORE D15 x 8.5 FROM BOTTOM	axis=[0, 0, 1], model_depth_mm=8.5, model_diameter_mm=15 全局入口坐标 XYZ: (-45, 90, 2); (45, 90, 2); (-45, 180, 2); (45, 180, 2)
B04-101-DECK M6x1-6H; FULL THREAD >=14; DRILL 17	axis=[1, 0, 0], model_depth_mm=17, model_diameter_mm=5 全局入口坐标 XYZ: (-110, 15, 8); (-110, 45, 8); (-110, 75, 8)
B04-101-DECK M6x1-6H; FULL THREAD >=14; DRILL 17	axis=[-1, 0, 0], model_depth_mm=17, model_diameter_mm=5 全局入口坐标 XYZ: (110, 15, 8); (110, 45, 8); (110, 75, 8)
B04-101-DECK D3.4 THRU	axis=[0, 0, 1], model_depth_mm=12, model_diameter_mm=3.4 全局入口坐标 XYZ: (-100, 35, 2); (100, 35, 2); (-100, 75, 2); (100, 75, 2); (-88, 185, 2); (88, 185, 2)
B04-101-DECK CSK D6.3 x 90 DEG FROM BOTTOM; M3x20 FLAT HEAD	axis=[0, 0, 1] 全局入口坐标 XYZ: (-100, 35, 2); (100, 35, 2); (-100, 75, 2); (100, 75, 2); (-88, 185, 2); (88, 185, 2)
B04-101-DECK M3x0.5-6H; FULL THREAD 6; DRILL 8	axis=[0, 0, -1], model_depth_mm=8, model_diameter_mm=2.5 全局入口坐标 XYZ: (-35, 3, 14); (35, 3, 14); (35, 43, 14); (-35, 43, 14)
B04-101-DECK M4x0.7-6H; FULL THREAD 7; CHASSIS BOND LUG	axis=[0, 0, -1], model_depth_mm=9, model_diameter_mm=3.3 全局入口坐标 XYZ: (90, -15, 14)
B04-101-DECK 36 x16 THRU WINDOW, R4; X43..79 Y-34...-18; CABLE DROP	全局入口坐标 XYZ: (61, -26, 2)
B04-102-CPLATE-L D6.6 THRU; M6x25 + WASHER	axis=[1, 0, 0], model_depth_mm=10, model_diameter_mm=6.6 全局入口坐标 XYZ: (-120, 15, 8); (-120, 45, 8); (-120, 75, 8)
B04-102-CPLATE-L D8.5 THRU; M8x25 + WASHER	axis=[1, 0, 0], model_depth_mm=10, model_diameter_mm=8.5 全局入口坐标 XYZ: (-120, 40, -101); (-120, 75, -101)
B04-102-CPLATE-L M6x1-6H THRU; CRADLE	axis=[1, 0, 0], model_depth_mm=10, model_diameter_mm=5 全局入口坐标 XYZ: (-120, -18, -70); (-120, -18, -95)
B04-102-CPLATE-R D6.6 THRU; M6x25 + WASHER	axis=[-1, 0, 0], model_depth_mm=10, model_diameter_mm=6.6 全局入口坐标 XYZ: (120, 15, 8); (120, 45, 8); (120, 75, 8)
B04-102-CPLATE-R D8.5 THRU; M8x25 + WASHER	axis=[-1, 0, 0], model_depth_mm=10, model_diameter_mm=8.5 全局入口坐标 XYZ: (120, 40, -101); (120, 75, -101)
B04-102-CPLATE-R M6x1-6H THRU; CRADLE	axis=[-1, 0, 0], model_depth_mm=10, model_diameter_mm=5 全局入口坐标 XYZ: (120, -18, -70); (120, -18, -95)
B04-103-BRIDGE M12x1.75-6H THRU; BREAK BOTH ENTRIES 0.5	axis=[0, 0, 1], model_depth_mm=22, model_diameter_mm=10.2 全局入口坐标 XYZ: (-85, 55, -112); (85, 55, -112)
B04-103-BRIDGE M8x1.25-6H; FULL THREAD >=15; DRILL 18	axis=[1, 0, 0], model_depth_mm=18, model_diameter_mm=6.8 全局入口坐标 XYZ: (-110, 40, -101); (-110, 75, -101)

manifest SHA256 ad065c979d159c1636a3b4f45636eb2d2e782b9553024c0c5ffa4498b730d9ef

相同标注已成组；全部入口坐标保留 / 本表不含采购包络的伪加工尺寸

特征 / 属性路径	数值、方向和限定说明 (长度默认 mm ; 其他单位以字段名为准)
B04-103-BRIDGE M8x1.25-6H; FULL THREAD >=15; DRILL 18	axis=[-1, 0, 0], model_depth_mm=18, model_diameter_mm=6.8 全局入口坐标 XYZ: (110, 40, -101); (110, 75, -101)
B04-104-FLANGE D8.5 THRU; M8x25 + WASHER	axis=[0, 0, 1], model_depth_mm=12, model_diameter_mm=8.5 全局入口坐标 XYZ: (-45, 90, 46); (45, 90, 46); (-45, 180, 46); (45, 180, 46)
B04-104-FLANGE M6x1-6H THRU; PCD120	axis=[0, 0, 1], model_depth_mm=12, model_diameter_mm=5 全局入口坐标 XYZ: (55.43277, 157.96101, 46); (22.96101, 190.43277, 46); (-22.96101, 190.43277, 46); (-55.43277, 157.96101, 46); (-55.43277, 112.03899, 46); (-22.96101, 79.56723, 46); (22.96101, 79.56723, 46); (55.43277, 112.03899, 46)
B04-105-POST-1 M8x1.25-6H THRU	axis=[0, 0, 1], model_depth_mm=32, model_diameter_mm=6.8 全局入口坐标 XYZ: (-45, 90, 14)
B04-105-POST-2 M8x1.25-6H THRU	axis=[0, 0, 1], model_depth_mm=32, model_diameter_mm=6.8 全局入口坐标 XYZ: (45, 90, 14)
B04-105-POST-3 M8x1.25-6H THRU	axis=[0, 0, 1], model_depth_mm=32, model_diameter_mm=6.8 全局入口坐标 XYZ: (-45, 180, 14)
B04-105-POST-4 M8x1.25-6H THRU	axis=[0, 0, 1], model_depth_mm=32, model_diameter_mm=6.8 全局入口坐标 XYZ: (45, 180, 14)
B04-107-PRESSPAD--1 M3x0.5-6H; FULL THREAD4; DRILL6; FOOT RETAINER	axis=[0, 0, 1], model_depth_mm=6, model_diameter_mm=2.5 全局入口坐标 XYZ: (-108, 55, -40); (-62, 55, -40)
B04-109-FOOT-CAGE--1--1 D3.4 THRU; M3x14, ENGAGEMENT4	axis=[0, 0, 1], model_depth_mm=10, model_diameter_mm=3.4 全局入口坐标 XYZ: (-108, 55, -50)
B04-109-FOOT-CAGE--1-1 D3.4 THRU; M3x14, ENGAGEMENT4	axis=[0, 0, 1], model_depth_mm=10, model_diameter_mm=3.4 全局入口坐标 XYZ: (-62, 55, -50)
B04-107-PRESSPAD-1 M3x0.5-6H; FULL THREAD4; DRILL6; FOOT RETAINER	axis=[0, 0, 1], model_depth_mm=6, model_diameter_mm=2.5 全局入口坐标 XYZ: (62, 55, -40); (108, 55, -40)
B04-109-FOOT-CAGE-1--1 D3.4 THRU; M3x14, ENGAGEMENT4	axis=[0, 0, 1], model_depth_mm=10, model_diameter_mm=3.4 全局入口坐标 XYZ: (62, 55, -50)
B04-109-FOOT-CAGE-1-1 D3.4 THRU; M3x14, ENGAGEMENT4	axis=[0, 0, 1], model_depth_mm=10, model_diameter_mm=3.4 全局入口坐标 XYZ: (108, 55, -50)
B04-201-CRADLE-L D6.6 THRU; M6x20 + WASHER	axis=[1, 0, 0], model_depth_mm=8, model_diameter_mm=6.6 全局入口坐标 XYZ: (-128, -18, -70); (-128, -18, -95)
B04-201-CRADLE-L M4x0.7-6H; FULL THREAD9; DRILL12	axis=[0, 0, -1], model_depth_mm=12, model_diameter_mm=3.3 全局入口坐标 XYZ: (-124, 140, -175); (-124, 250, -175)
B04-201-CRADLE-R D6.6 THRU; M6x20 + WASHER	axis=[-1, 0, 0], model_depth_mm=8, model_diameter_mm=6.6 全局入口坐标 XYZ: (128, -18, -70); (128, -18, -95)

manifest SHA256 ad065c979d159c1636a3b4f45636eb2d2e782b9553024c0c5ffa4498b730d9ef

相同标注已成组；全部入口坐标保留 / 本表不含采购包络的伪加工尺寸

特征 / 属性路径	数值、方向和限定说明 (长度默认 mm ; 其他单位以字段名为准)
B04-201-CRADLE-R M4x0.7-6H; FULL THREAD9; DRILL12	axis=[0, 0, -1], model_depth_mm=12, model_diameter_mm=3.3 全局入口坐标 XYZ: (124, 140, -175); (124, 250, -175)
B04-202-TRAY D4.5 THRU; M4x12	axis=[0, 0, 1], model_depth_mm=5, model_diameter_mm=4.5 全局入口坐标 XYZ: (-124, 140, -175); (-124, 250, -175); (124, 140, -175); (124, 250, -175)
B04-202-TRAY M4x0.7-6H THRU; STOP M4x16	axis=[0, 0, 1], model_depth_mm=5, model_diameter_mm=3.3 全局入口坐标 XYZ: (-100, 112, -175); (100, 112, -175); (-100, 278, -175); (100, 278, -175)
B04-202-TRAY 4 x24 THRU SLOT, R1.5; 20mm RETAINING STRAP	全局入口坐标 XYZ: (-108, 187, -175); (108, 187, -175)
B04-203-STOP-REAR D4.5 THRU; M4x16	axis=[0, 0, 1], model_depth_mm=12, model_diameter_mm=4.5 全局入口坐标 XYZ: (-100, 112, -170); (100, 112, -170)
B04-203-STOP-FRONT D4.5 THRU; M4x16	axis=[0, 0, 1], model_depth_mm=12, model_diameter_mm=4.5 全局入口坐标 XYZ: (-100, 278, -170); (100, 278, -170)
B04-301-COVER-REAR D3.4 THRU; M3x8 + WASHER	axis=[0, 0, 1], model_depth_mm=3, model_diameter_mm=3.4 全局入口坐标 XYZ: (-100, 35, 49); (100, 35, 49)
B04-301-COVER-FRONT D3.4 THRU; M3x8 + WASHER	axis=[0, 0, 1], model_depth_mm=3, model_diameter_mm=3.4 全局入口坐标 XYZ: (-100, 75, 49); (100, 75, 49); (-88, 185, 49); (88, 185, 49)